



Advanced Subnetting & Problem-Solving Techniques (IPv4)

Focus

Subnetting is the **single most dominant IPv4 topic in GATE**, and success depends less on binary conversion and more on **pattern recognition**.

This document centers on **AD Rule 2.0**, a shortcut framework that replaces full bitwise ANDing with **selective octet analysis, decimal weight logic, and binary indexing**.

Key outcomes:

- Determine subnet membership in seconds
- Infer valid subnet masks from a DBA
- Identify exact subnets, hosts, and broadcast addresses without ranges
- Know when shortcuts work—and when they *must* be abandoned

1 Fundamental Principles of Subnetting Analysis

1.1 DBA–Mask Relationship (High-Yield Rule)

A **Direct Broadcast Address (DBA)** has:

- All 1s in the Host ID (HID)

Compatibility Condition

To test whether a subnet mask can produce a given DBA:

[
 \text{Mask Host Bits (zeros)} \leq \text{Continuous trailing 1s in DBA}
]

✓ Count **continuous** 1s from the **rightmost side only**

✗ Ignore non-continuous 1s (they do not form a valid host field)

This rule alone eliminates most wrong options in GATE MCQs.

1.2 Binary Weighting (Mental Math Backbone)

Instead of full binary conversion, always think in **weights**:

Bit 128 64 32 16 8 4 2 1

Only check the weights where:

- The subnet mask has **1s**
- The IP address has matching weights

1.3 Subnet Mask Composition (Conceptual Separation)

Component	Meaning
Network ID (NID)	Fixed by class (A=8, B=16, C=24 bits)
Subnet ID (SID)	Extra 1s beyond default mask
Host ID (HID)	Zeros in subnet mask

Golden Rule:

Subnetting changes only the Host part—not the class boundary.

2 AD Rule 2.0 – The Speed Framework

AD Rule 2.0 eliminates full ANDing by isolating the **interesting octet**.

What is the “Interesting Octet”?

The octet where the subnet mask is **neither 255 nor 0**.

Only **this octet** determines:

- Subnet membership
- Subnet number
- Host position

2.1 Determining Subnet Membership (Same Subnet?)

Instead of ANDing entire addresses:

Procedure

1. Identify SID bit positions in the interesting octet
2. Extract **only those bits** from each IP
3. If SID bits match → same subnet
4. If not → different subnet

- ✓ Works in ~10 seconds
- ✓ Extremely reliable for multi-IP comparison questions

2.2 Finding Subnet Numbers & Subnet IDs

Subnet numbering follows **zero-based binary indexing**.

Binary SID	Decimal	Subnet Number
000	0	1st subnet
001	1	2nd subnet
010	2	3rd subnet
n	n	(n+1) th subnet

Rule

[
 $\text{Binary of } (n-1) = \text{SID of } n^{\text{th}} \text{ subnet}$
]

To get:

- **Subnet ID** → set all HID bits to 0
- **DBA** → set all HID bits to 1

③ Specialized Case Studies (Exam-Style Reasoning)

3.1 Class B + DBA Inference

Given: DBA = 144.1.95.255 (Class B)

- Host portion = last 2 octets
- Binary:
 - 95 → 01011111
 - 255 → 11111111
- Continuous trailing 1s = 13

✓ Therefore:

[
\text{Host bits} \leq 13
]

- ✓ Any mask with ≤ 13 zeros is valid
- ✓ Faster than testing each mask manually

3.2 Non-Continuous Subnet Masks (Trap Zone)

Example mask octet: 198 → 11000110

⚠ Why shortcuts fail:

- Ones are **not contiguous**
- Host bits are scattered

✓ Only safe method:

- Perform **manual bitwise reasoning**
- Explicitly verify if setting host bits to 1 recreates the DBA

Exam Insight:

If mask bits are non-continuous, **do not use AD shortcuts.**

3.3 Class C – Finding a Specific Host

Problem:

Find **3rd host of the 2nd subnet**, Class C, SID = 3 bits

Steps

1. 2nd subnet → SID = 1 → 001
2. Host bits = 5

3. 3rd host → HID = 00011

4. Combine:

- NID (24 bits)
- SID = 001
- HID = 00011

- ✓ No ranges
- ✓ No ANDing
- ✓ Direct construction

4 Universal Address Construction Rules

Target	Bit Pattern
Subnet ID	HID = all 0s
First Host	HID = 000...001
Last Host	HID = 111...110
DBA	HID = all 1s
Subnet Number	Decimal(SID) + 1

5 Practical GATE Insights (Scoring Edge)

- ♦ **Always identify the class first** → fixes NID boundary
- ♦ **Subnet ID ≠ Subnet Number** (most common mistake)
- ♦ Masks with 255 or 0 octets need **zero analysis**
- ♦ Analyze only octets between 0 and 255
- ♦ Continuous masks → shortcuts valid
- ♦ Non-continuous masks → revert to fundamentals